PATIENT DATA ANALYSIS USING KNIME

Health Care Data Analytics Project

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PROJECT INTRODUCTION AND PROBLEM STATEMENT:

The Healthcare System project simulates patient data to generate insights for hospital management. It uses three datasets Patient Profile, Treatment Records, and Visit/Follow-ups. The project identifies patients at risk, analyses treatment outcomes, and studies the impact of follow-up compliance on recovery. This helps demonstrate how data analysis can improve healthcare decisions and patient care.

PROJECT OBJECTIVES:

The primary objectives are to perform fundamental data manipulation tasks on the hospital datasets to facilitate data-driven decision making. The core tasks include:

- **Filtering** to identify specific patient segments (e.g. patients who missed follow-up visits, whose treatment outcomes are poor, filtering by age group etc).
- **Grouping and Aggregation** to calculate key metrics, such as average recovery rate, compare treatment outcomes etc.
- Joining multiple related tables using the common key **Patient ID** for easier analysis

DATA DESCRIPTION AND SIMULATION:

This project uses **three related datasets**, connected through a common key **Patient ID**. These datasets together provide a complete view of patient health, treatment progress, and follow-up compliance.

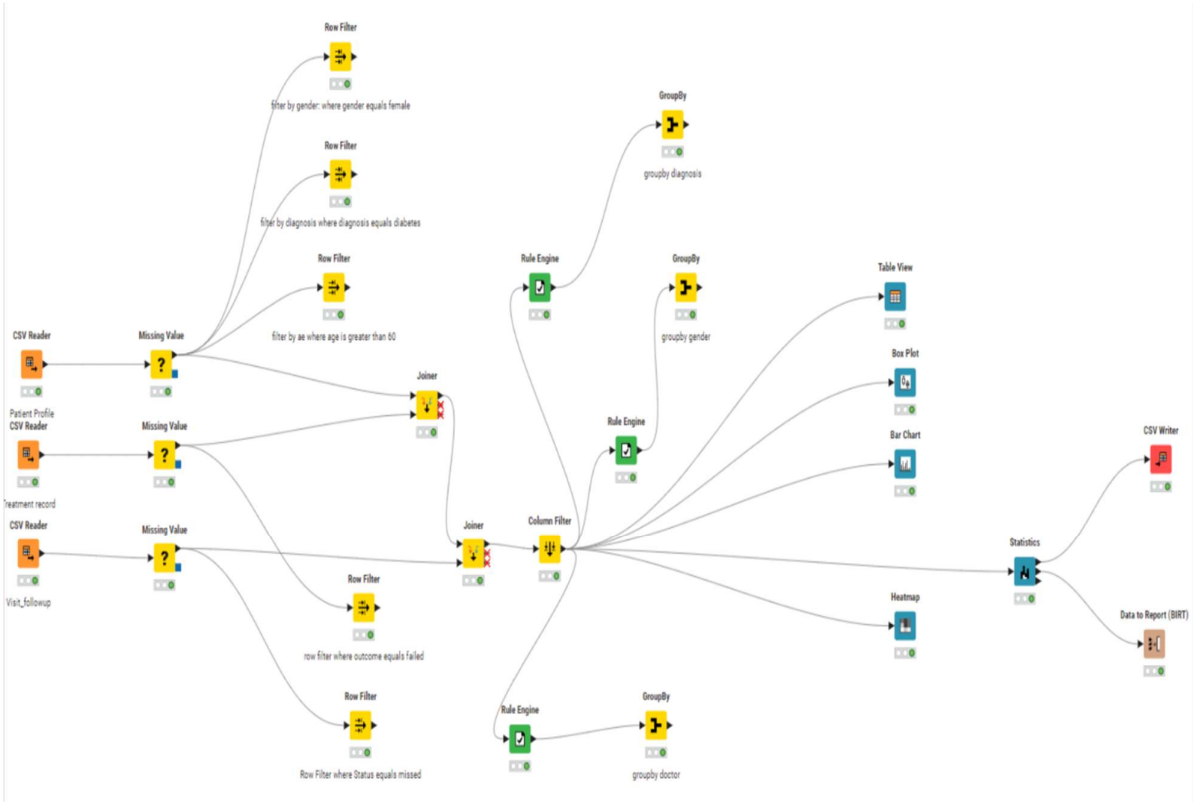
- **Patient Profile Dataset:**
  Contains core demographic and medical details of each patient, including **Patient ID**, **Name**, **Age**, **Gender**, **Diagnosis** This dataset forms the foundation for identifying and categorizing patients based on their health conditions.
- **Treatment Dataset:**

  This dataset stores detailed information about the treatments provided to each patient.
  It includes attributes like **Patient ID**, **Treatment Procedure**, **Outcome**, **Start Date**
  It helps analyse how different treatments are administered and their impact on patient recovery.

- **Follow-up Dataset**:

  This dataset records the follow-up and visit details of patients after treatment.
  It contains **Patient ID**, **Visit Date**, **Doctor** , **Follow-up Status**
  It is used to track whether patients attend their scheduled follow-ups and which doctor they visited

KNIME WORKFLOW :



WORKFLOW AND METHODOLOGY:

The project requires simulating or importing the hospital datasets.

KNIME NODE	PURPOSE	INPUT/ CONFIGURATION	EXPECTED OUTPUT
Table Creator	Table Creator (x3)[2]	To manually simulate the three required datasets: patient profiles, treatment records, Visit follow-up	Configure schema and enter sample data for each dataset3.Three separate table containing the simulated data columns
Table Creator	CSV Creator or CSV Reader	Alternative for generating or importing larger, more realistic datasets.	Point to a local CSV file or configure data generation parameters. One or more output data tables ready for processing.

OUTPUT:

Table: patient profile

#	RowID	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String
1	Row0	1	Sarah Little	43	Male	Flu
2	Row1	2	Willie Acevedo	10	Other	Arthritis
3	Row2	3	Melissa Stone	55	Male	Flu
4	Row3	4	Julie Rogers	63	Female	COVID-19
5	Row4	5	Ashley Ruiz	51	Male	Flu
6	Row5	6	Steven Keith	82	Female	Flu
7	Row6	7	Loretta Russell	10	Other	Arthritis

Table: Treatment records

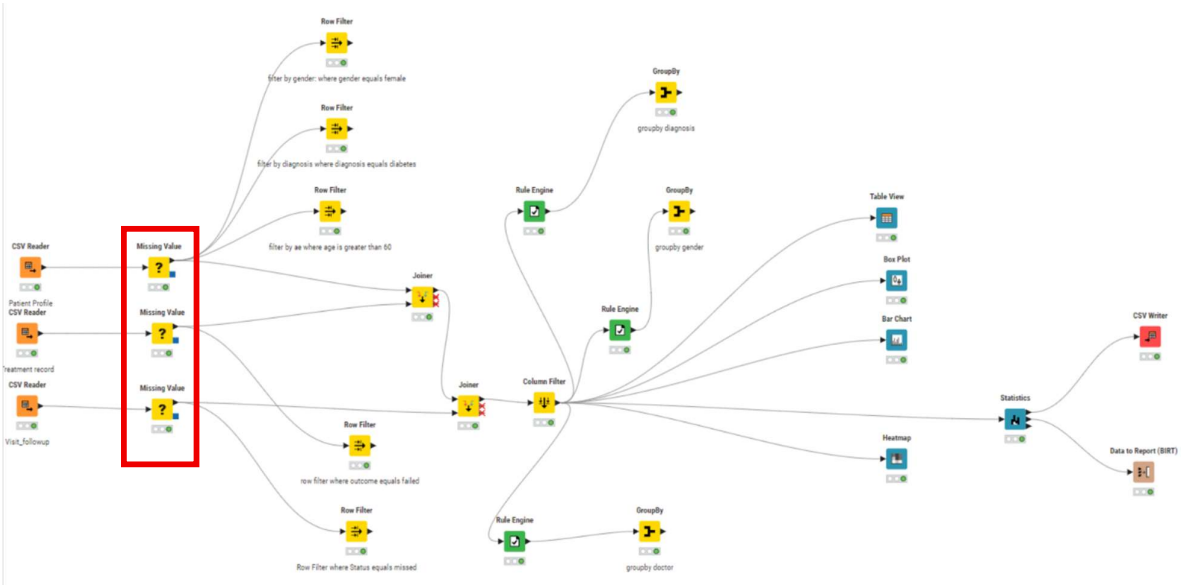
RowID	Patient ID Number (Integer)	Treatment/Procedure String	Outcome String	Date String
Row0	1	Chemotherapy	Ongoing	20-09-2025
Row1	2	Vaccination	?	31-08-2025
Row2	2	Counseling	Successful	12-07-2025
Row3	2	Surgery	Failed	24-05-2024
Row4	3	Medication	Ongoing	14-10-2024
Row5	3	Counseling	Successful	11-05-2025
Row6	3	Surgery	Failed	23-03-2024
Row7	4	Chemotherapy	Successful	25-01-2024

Table: Visit followup

RowID	Patient ID ↓ Number (Integer)	Visit Date String	Doctor String	Status String
Row0	1	09-07-2024	Chloe Fox	Completed
Row1	1	14-08-2024	Ryan Vincent	Completed
Row2	1	25-03-2025	Alexandria Hodge	?
Row3	2	10-01-2025	Cheryl Mason	Completed
Row4	2	28-12-2023	Robert Patrick	Missed
Row5	3	08-12-2024	Christopher Haas	Completed
Row6	3	10-02-2025	Christina Moore	Completed
Row7	4	18-05-2025	Anthony Schmidt	Missed

FIRST CHECKS AND PRE-PROCESSING:

Handle missing values using the Missing Value node.



After applying the missing values node on all the three datasets the missing values of the data will get updated

Updated table : patient profile

RowID	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String
Row0	1	Sarah Little	43	Male	Flu
Row1	2	Willie Acevedo	10	Other	Arthritis
Row2	3	Melissa Stone	55	Male	Flu
Row3	4	Julie Rogers	63	Female	COVID-19
Row4	5	Ashley Ruiz	51	Male	Flu
Row5	6	Steven Keith	82	Female	Flu
Row6	7	Loretta Russell	10	Other	Arthritis

Updated table: treatment record

RowID	Patient ID ↓ Number (Integer)	Treatment/Procedure String	Outcome String	Date String
Row0	1	Chemotherapy	Ongoing	20-09-2025
Row1	2	Vaccination	Ongoing	31-08-2025
Row2	2	Counseling	Successful	12-07-2025
Row3	2	Surgery	Failed	24-05-2024
Row4	3	Medication	Ongoing	14-10-2024
Row5	3	Counseling	Successful	11-05-2025
Row6	3	Surgery	Failed	23-03-2024
Row7	4	Chemotherapy	Successful	25-01-2024

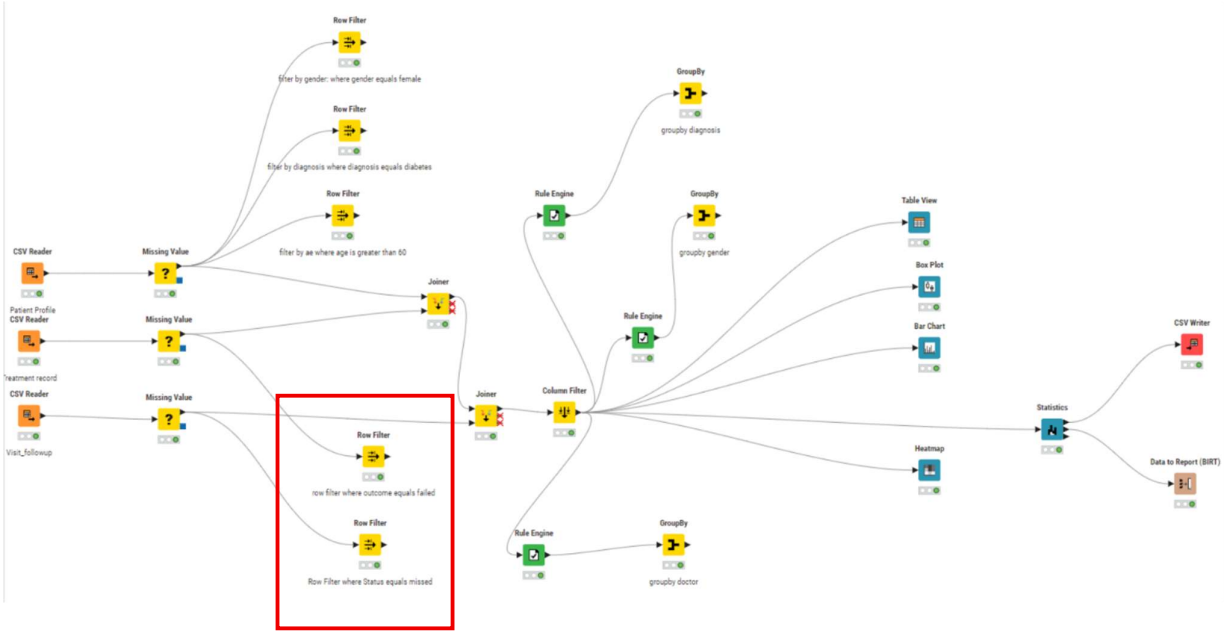
Updated table: Visit followup

RowID	Patient ID Number (Integer)	Visit Date ↓ String	Doctor String	Status String
Row0	1	09-07-2024	Chloe Fox	Completed
Row1	1	14-08-2024	Ryan Vincent	Completed
Row2	1	25-03-2025	Alexandria Hodge	Missed
Row3	2	10-01-2025	Cheryl Mason	Completed
Row4	2	28-12-2023	Robert Patrick	Missed
Row5	3	08-12-2024	Christopher Haas	Completed
Row6	3	10-02-2025	Christina Moore	Completed
Row7	4	18-05-2025	Anthony Schmidt	Missed

FILTERING TASK :

This stage ensures data quality and proper data types.

KNIME NODE	PURPOSE	INPUT/ CONFIGURATION	EXPECTED OUTPUT
Row Filter	Row Filter	Remove/filters out rows that contain invalid or clearly erroneous data points	A table containing only valid and clean records



OUTPUT:

Row Filter : Gender = 'Female'

R...	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String
Row3	4	Julie Rogers	63	Female	COVID-19
Row5	6	Steven Keith	82	Female	Flu
Row7	8	Taylor Coleman	83	Female	Diabetes
Row11	12	Cindy Parker	81	Female	Hypertension
Row12	13	Mrs. Joann Allen DDS	66	Female	Asthma

Row Filter : Diagnosis = 'Diabetes'

RowID	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String
Row7	8	Taylor Coleman	83	Female	Diabetes
Row13	14	Jeff Yoder	40	Female	Diabetes
Row14	15	William Collins	76	Female	Diabetes
Row32	33	Benjamin Myers	65	Female	Diabetes
Row35	36	Derek Chavez	90	Female	Diabetes

Row Filter : Age greater than 60

RowID	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String
Row3	4	Julie Rogers	63	Female	COVID-19
Row5	6	Steven Keith	82	Female	Flu
Row7	8	Taylor Coleman	83	Female	Diabetes
Row8	9	Kenneth Fields	83	Other	Flu
Row10	11	Jennifer Brady	78	Male	Hypertension

Row Filter : Outcome = 'failed'

RowID	Patient ID Number (Integer)	Treatment/Procedure String	Outcome String	Date String
Row3	2	Surgery	Failed	24-05-2024
Row6	3	Surgery	Failed	23-03-2024
Row8	5	Physiotherapy	Failed	31-03-2024
Row9	6	Vaccination	Failed	28-03-2025
Row12	8	Surgery	Failed	08-07-2024

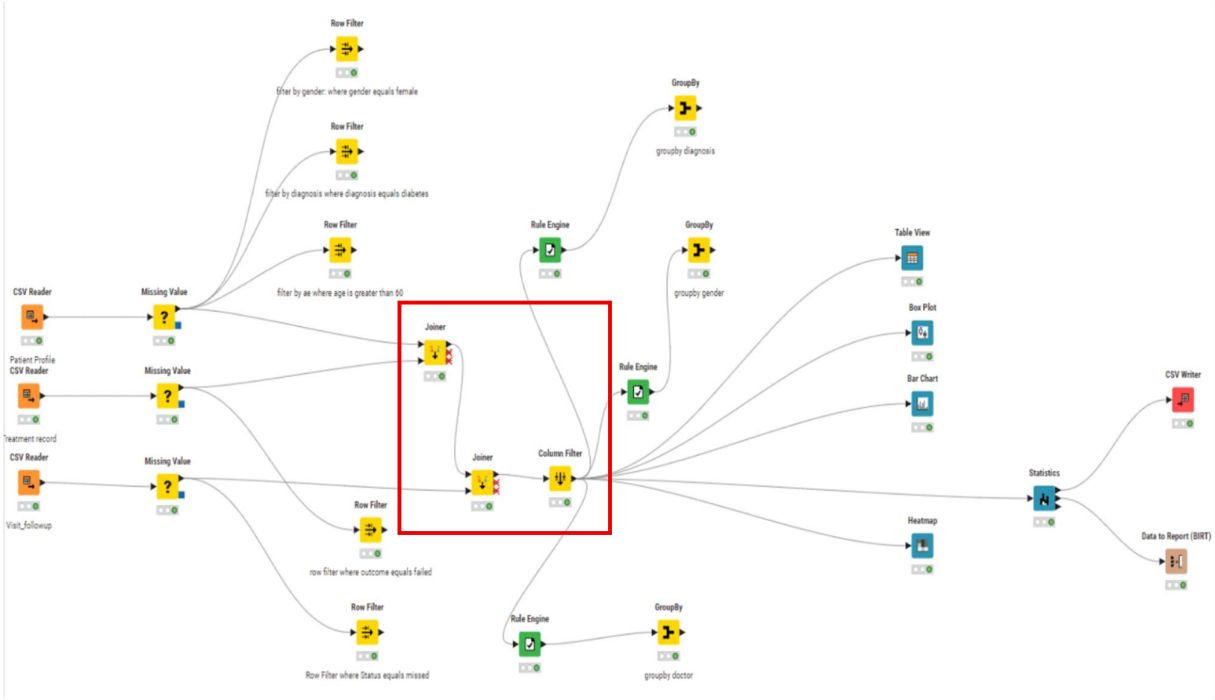
Row Filter : Status = 'Missed'

RowID	Patient ID Number (Integer)	Visit Date String	Doctor String	Status String
Row2	1	25-03-2025	Alexandria Hodge	Missed
Row4	2	28-12-2023	Robert Patrick	Missed
Row7	4	18-05-2025	Anthony Schmidt	Missed
Row10	5	26-05-2024	Julie Johnson	Missed
Row13	6	03-10-2023	Jody Williams	Missed

JOINING DATASETS :

The join operation links the separate employee records based on patient ID.

KNIME NODE	PURPOSE	INPUT/ CONFIGURATION	EXPECTED OUTPUT
Joiner	Joiner	Input 1: Patient profile; Input 2: Treatment record After joining these two tables join it with visit followup table	A common table joined on patient id



OUTPUT:

Joining all three tables into a single table using joiner on patient id

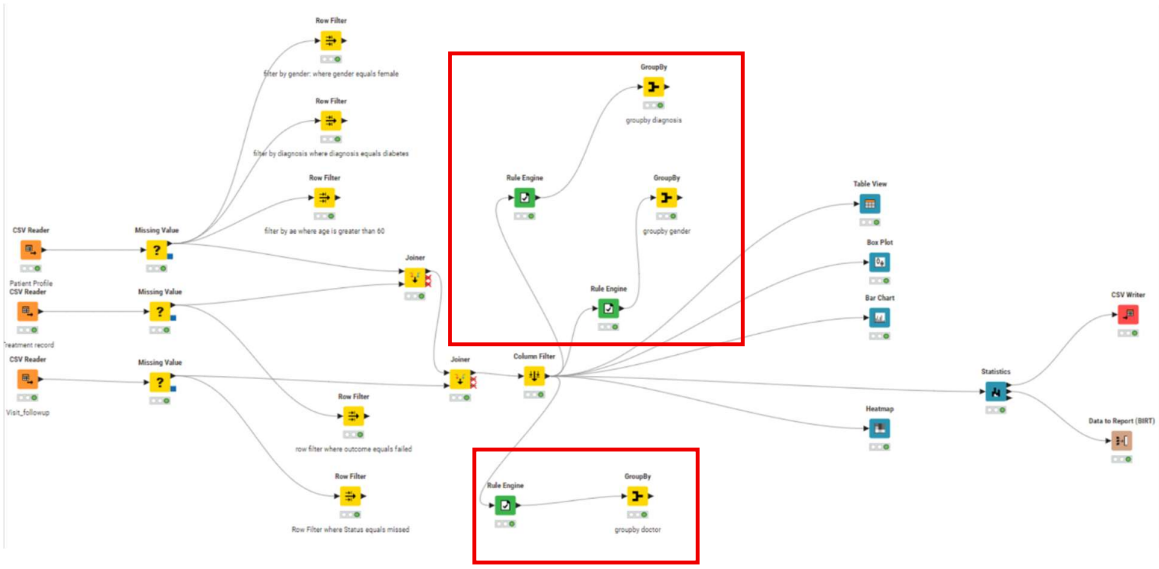
RowID	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String	Treatment/P... String	Outcome String	Date String	Visit Date String	Doctor String	Status String
Row0_I	1	Sarah Little	43	Male	Flu	Chemotherapy	Ongoing	20-09-2025	09-07-2024	Chloe Fox	Completed
Row0_I	1	Sarah Little	43	Male	Flu	Chemotherapy	Ongoing	20-09-2025	14-08-2024	Ryan Vincent	Completed
Row0_I	1	Sarah Little	43	Male	Flu	Chemotherapy	Ongoing	20-09-2025	25-03-2025	Alexandria Hodge	Missed
Row1_I	2	Willie Acevedo	10	Other	Arthritis	Vaccination	Ongoing	31-08-2025	10-01-2025	Cheryl Mason	Completed
Row1_I	2	Willie Acevedo	10	Other	Arthritis	Vaccination	Ongoing	31-08-2025	28-12-2023	Robert Patrick	Missed
Row1_I	2	Willie Acevedo	10	Other	Arthritis	Counseling	Successful	12-07-2025	10-01-2025	Cheryl Mason	Completed
Row1_I	2	Willie Acevedo	10	Other	Arthritis	Counseling	Successful	12-07-2025	28-12-2023	Robert Patrick	Missed

GROUPING AND AGGREGATION :

The GroupBy node is used to summarize the data and calculate average metrics by various categories.

KNIME NODE	PURPOSE	INPUT/ CONFIGURATION	EXPECTED OUTPUT
Groupby	Groupby	Calculate average recovery rates by diagnosis	A small summary table having columns diagnosis and mean(prediction) recovery rates

Groupby	Groupby	Calculate treatment success rate by doctor	A small summary table having columns gender and their recovery rates mean(prediction)
Groupby	Groupby	Comparitive analysis by gender	A small recovery table having columns doctor and treatment recovery rates mean(prediction)



OUTPUT:

Since the recovery rate is a categorical value we used rule engine node to convert it into numerical (recovered = 1 else = 0) after that we found the mean of that value.

GroupBy : Avg recovery rate by diagnosis

RowID	Diagnosis *String*	Mean(prediction) *Number (Float)*
Row0	Arthritis	0.103
Row1	Asthma	0.311
Row2	COVID-19	0.493
Row3	Diabetes	0.092
Row4	Flu	0.148
Row5	Heart Disease	0.269
Row6	Hypertension	0.242

GroupBy : Avg recovery rate by gender

RowID	Gender *String*	Mean(prediction) *Number (Float)*
Row0	Female	0.172
Row1	Male	0.2
Row2	Other	0.363

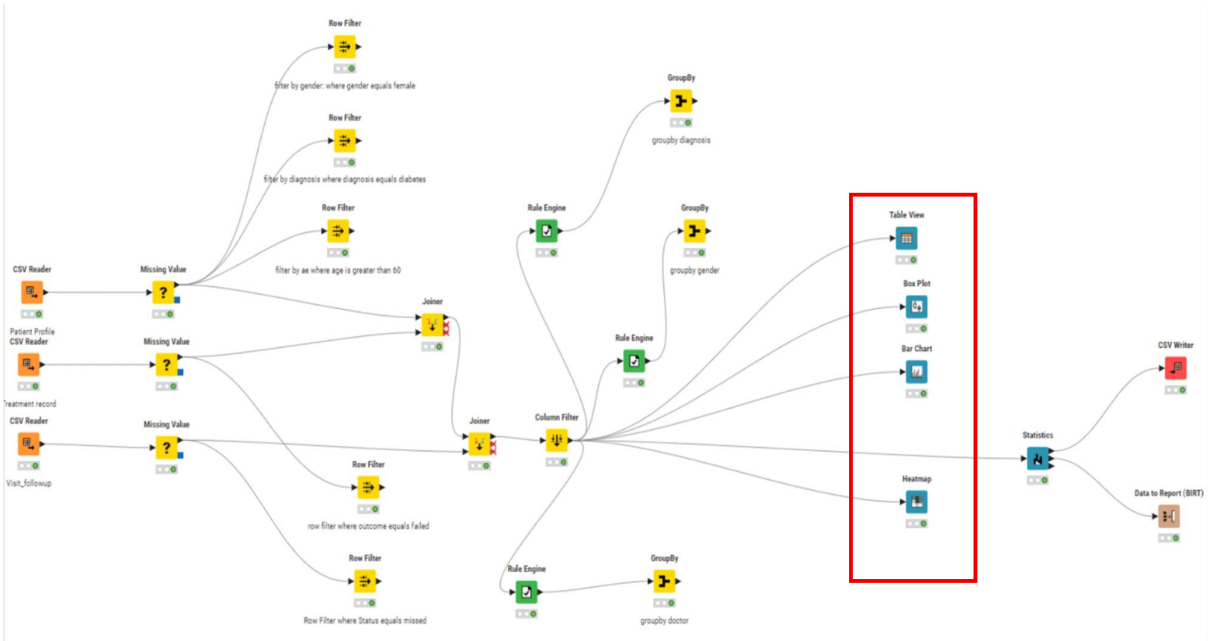
GroupBy : Avg recovery rate by Doctor

RowID	Doctor *String*	Mean(prediction) *Number (Float)*
Row0	Adam Caldwell	1
Row1	Albert Moore	0
Row2	Albert Smith	0
Row3	Alec Hamilton	0
Row4	Alexandria Hodge	0
Row5	Alicia Hall	0
Row6	Alison Morgan	0.333

EXPLORATION AND REPORTING :

The final stage involves visualizing and exporting the results.

KNIME NODE	PURPOSE	INPUT/ CONFIGURATION	EXPECTED OUTPUT
Bar chart	Plot a Bar chart	Treatment procedure and concurrent count	Bar chart is plotted
Box plot	Plot a Box plot	Age and gender	Box plot is plotted
Heat map	Plot a Heat map	Diagnosis and outcome	Heat map is plotted
Table view	View in table form	The merged dataset	Table is created

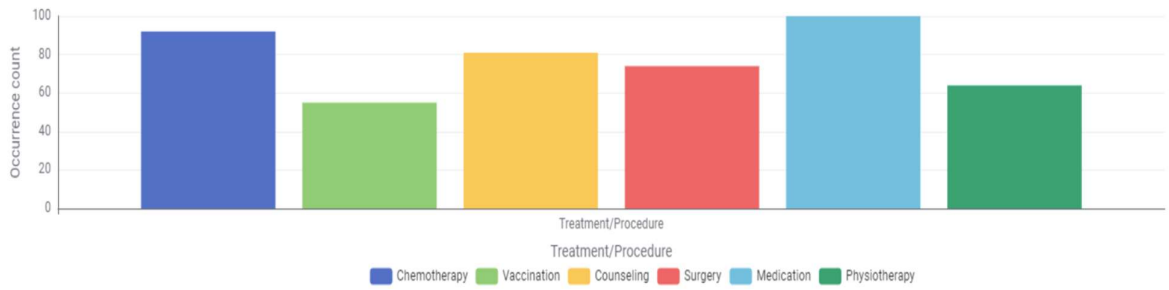


OUTPUT:

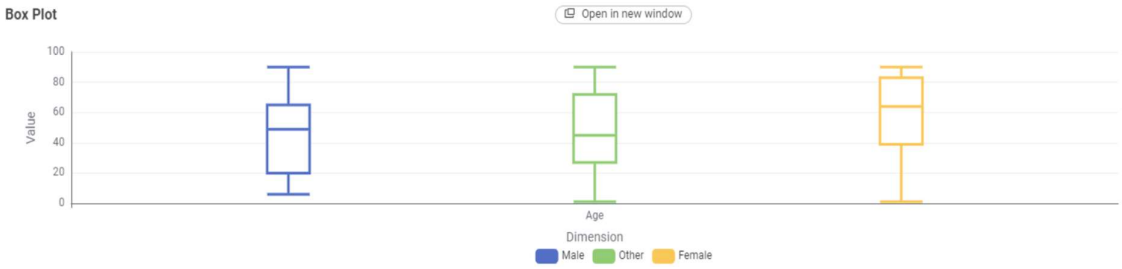
Table view

RowID	Patient ID Number (Integer)	Name String	Age Number (Integer)	Gender String	Diagnosis String	Treatment/Pr... String	Outcome String	Date String	Visit Date String	Doctor String	Status String
Row0_	1	Sarah Little	43	Male	Flu	Chemotherapy	Ongoing	20-09-2025	09-07-2024	Chloe Fox	Completed
Row0_	1	Sarah Little	43	Male	Flu	Chemotherapy	Ongoing	20-09-2025	14-08-2024	Ryan Vincent	Completed
Row0_	1	Sarah Little	43	Male	Flu	Chemotherapy	Ongoing	20-09-2025	25-03-2025	Alexandria Hodge	Missed
Row1_	2	Willie Acevedo	10	Other	Arthritis	Vaccination	Ongoing	31-08-2025	10-01-2025	Cheryl Mason	Completed

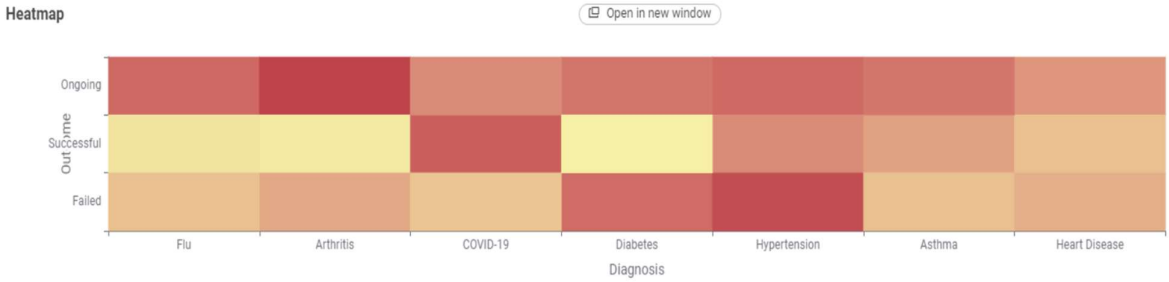
Bar chart



Box plot



Heat map



WRITING THE DATA AND MAKING SUMMARY

Use excel writer/CSV writer to save the final output and summary table

Merged table:

	A	B	C	D	E	F	G	H	I	J	K
1	Patient ID	Name	Age	Gender	Diagnosis	Treatment	Outcome	Date	Visit Date	Doctor	Status
2	1	Sarah Littl	43	Male	Flu	Chemothe	Ongoing	20-09-2025	09-07-2024	Chloe Fox	Completed
3	1	Sarah Littl	43	Male	Flu	Chemothe	Ongoing	20-09-2025	14-08-2024	Ryan Vince	Completed
4	1	Sarah Littl	43	Male	Flu	Chemothe	Ongoing	20-09-2025	25-03-2025	Alexandria	Missed
5	2	Willie Acev	10	Other	Arthritis	Vaccinatio	Ongoing	31-08-2025	10-01-2025	Cheryl Ma:	Completed
6	2	Willie Acev	10	Other	Arthritis	Vaccinatio	Ongoing	31-08-2025	28-12-2023	Robert Pat	Missed
7	2	Willie Acev	10	Other	Arthritis	Counseling	Successful	12-07-2025	10-01-2025	Cheryl Ma:	Completed
8	2	Willie Acev	10	Other	Arthritis	Counseling	Successful	12-07-2025	28-12-2023	Robert Pat	Missed
9	2	Willie Acev	10	Other	Arthritis	Surgery	Failed	24-05-2024	10-01-2025	Cheryl Ma:	Completed
10	2	Willie Acev	10	Other	Arthritis	Surgery	Failed	24-05-2024	28-12-2023	Robert Pat	Missed
11	3	Melissa St	55	Male	Flu	Medicatio	Ongoing	14-10-2024	08-12-2024	Christophe	Completed
12	3	Melissa St	55	Male	Flu	Medicatio	Ongoing	14-10-2024	10-02-2025	Christina N	Completed
13	3	Melissa St	55	Male	Flu	Counseling	Successful	11-05-2025	08-12-2024	Christophe	Completed
14	3	Melissa St	55	Male	Flu	Counseling	Successful	11-05-2025	10-02-2025	Christina N	Completed
15	3	Melissa St	55	Male	Flu	Surgery	Failed	23-03-2024	08-12-2024	Christophe	Completed
16	3	Melissa St	55	Male	Flu	Surgery	Failed	23-03-2024	10-02-2025	Christina N	Completed
17	4	Julie Roger	63	Female	COVID-19	Chemothe	Successful	25-01-2024	18-05-2025	Anthony S	Missed
18	5	Ashley Rui:	51	Male	Flu	Physiother	Failed	31-03-2024	16-12-2024	Christophe	Completed

Summary table:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Column	Min	Max	Mean	Std. deviat	Variance	Skewness	Kurtosis	Overall sur	No. missin	No. NaNs	No. +?s	No. -?s	Median	Row count	Histogram
2	Patient ID	1	100	49.96781	27.61932	762.8269	0.099931	-1.06823	23285	0	0	0	0		466	
3	Age	1	90	50.3133	27.09713	734.2543	-0.13401	-1.21979	23446	0	0	0	0		466	
4																

TARGET INSIGHTS AND BUSINESS QUESTIONS :

The data manipulation workflow is designed to answer key business questions and generate actionable insights:

- **How can hospitals improve patient recovery outcomes through data-driven monitoring?**
  - By analysing treatment success rates and follow-up compliance, hospital management can identify which patient segments (e.g., age group, diagnosis type, or doctor) require more attention or different treatment strategies.
- **Which doctors and departments demonstrate the highest treatment efficiency and patient satisfaction?**
  - Measuring and comparing success rates by doctor and diagnosis helps management allocate resources, recognize high-performing staff, and identify where additional training or support may be needed.
- **Can predictive insights reduce missed follow-up visits and prevent patient dropouts?**
  - By tracking patterns in missed appointments and linking them with treatment outcomes, the hospital can develop targeted reminders or engagement strategies to ensure better patient adherence and overall care quality.

CONCLUSION:

This project demonstrates how data analytics can play a crucial role in improving healthcare management. By integrating and analyzing patient, treatment, and follow-up data using KNIME, meaningful insights such as recovery trends, doctor performance, and patient compliance were uncovered. These insights highlight the potential of data-driven decision-making in enhancing hospital efficiency, optimizing treatment plans, and improving overall patient outcomes.